



First Week of Development

Module 1

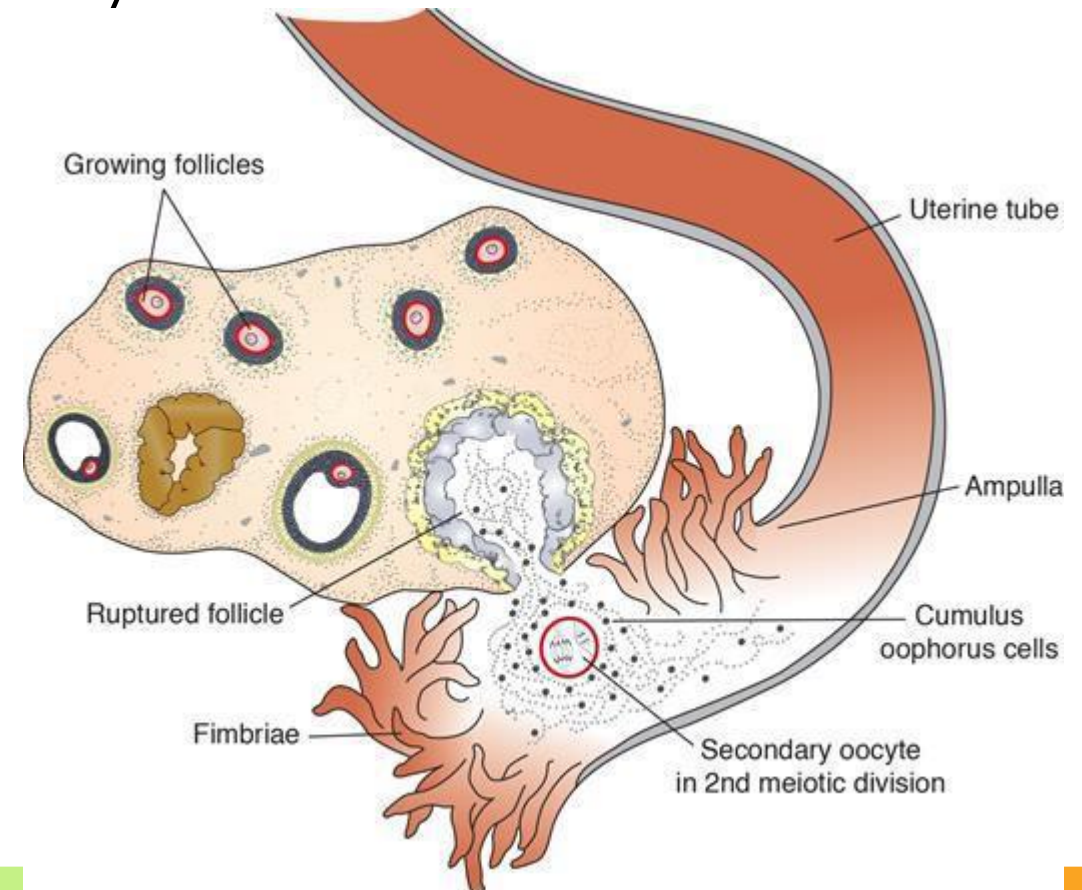
- Langman's Medical Embryology, 15th edition, pp. 34-49
- Helpful videos on YouTube:
- <https://www.youtube.com/watch?v=EwTZ1fypivg>
- <https://www.youtube.com/watch?v=bldJOiXpp9g>
- <https://www.youtube.com/watch?v=3AOoikTEfeo>
- <https://www.youtube.com/watch?v=IGLexQR9xGs>
- <https://www.youtube.com/watch?v=yXUv4MPuNTA>
- <https://www.youtube.com/watch?v=VoaxRbve42o>

LEARNING OBJECTIVES

1. List the events occurring during the first week of human development.
2. Define ovulation.
3. Describe the process of fertilization of the female gamete.
4. List the results of fertilization.
5. Describe cleavage.
6. List the abnormal sites of fertilization.

OVULATION

- Ovulation is defined as the process of extrusion of oocyte from a mature graffian follicle in the ovary into the pelvic cavity.
- Mittelschmerz-is the slight abdominal pain
- experienced at the time of ovulation.



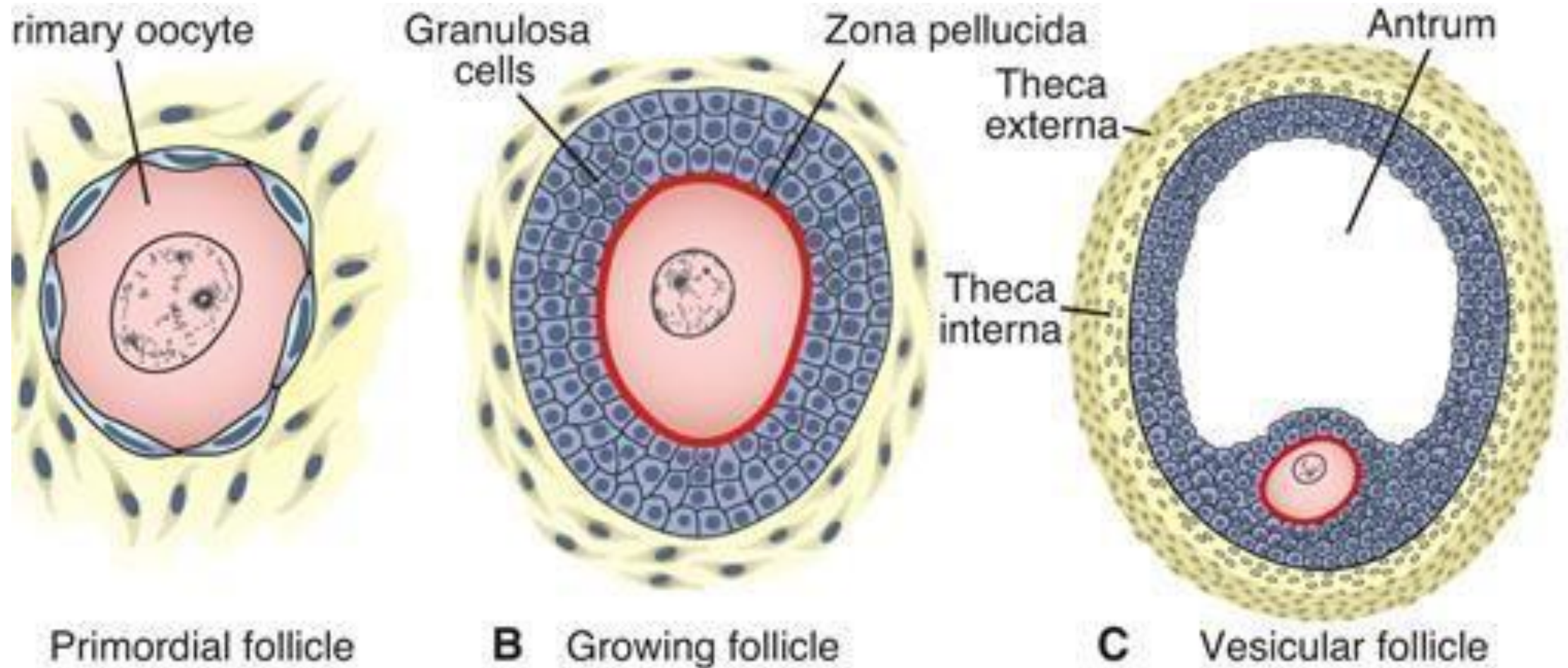
OVARIAN CYCLES

- At puberty, the female begins to undergo regular monthly cycles. These **sexual cycles** are controlled by the hypothalamus. **Gonadotropin-releasing hormone (GnRH)**, produced by the hypothalamus, acts on cells of the anterior lobe (adenohypophysis) of the pituitary gland, which in turn secretes **gonadotropins**. These hormones, **follicle-stimulating hormone (FSH)**, and **luteinizing hormone (LH)**, stimulate and control cyclic changes in the ovary.

OVARIAN CYCLES

- At the beginning of each cycle 15 to 20 primary follicles begin to grow under the influence of FSH.
- Only one of these follicles reaches full maturity, and only one oocyte is discharged; the others degenerate and become atretic.
- FSH also stimulates maturation of **follicular (granulosa)** cells surrounding the oocyte.
- Follicular cells are responsible for the secretion of Estrogen.

DEVELOPMENT OF OVARIAN FOLLICLES



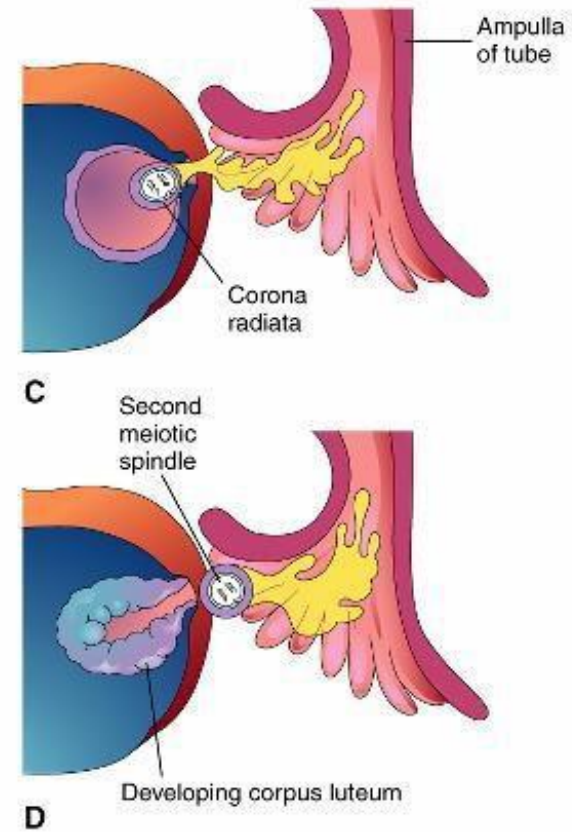
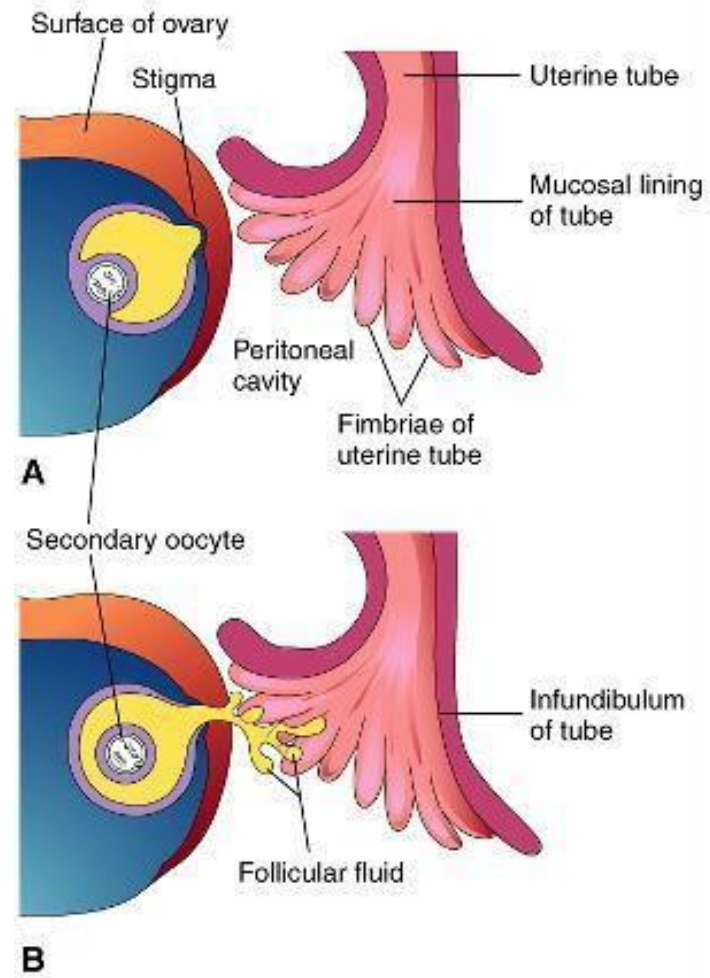
LH SURGE AND OVULATION

At midcycle, there is an LH surge that:

- Elevates concentrations of maturation-promoting factor, causing oocytes to complete meiosis I and initiate meiosis II
- Stimulates production of progesterone by follicular stromal cells (luteinization)
- Causes follicular rupture and ovulation

OVULATION

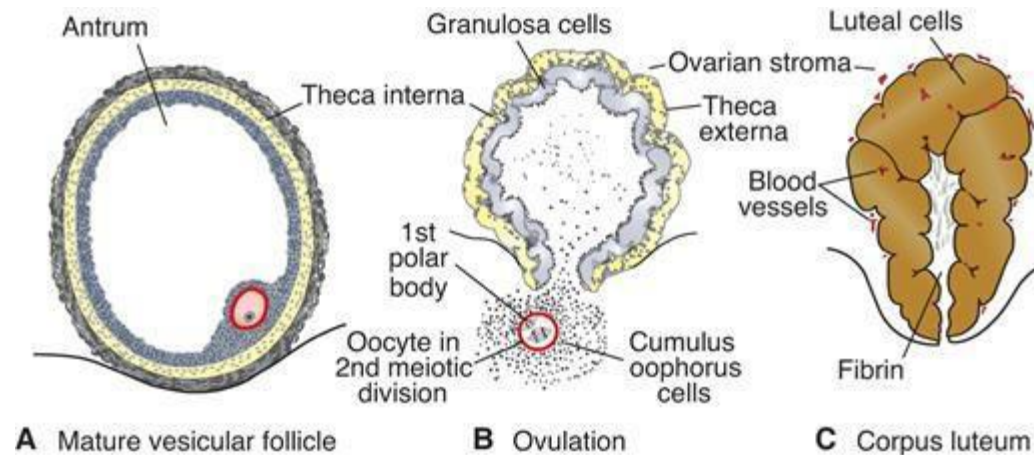
- Meiosis II is also initiated, but the oocyte is arrested in metaphase.
- The surface of the ovary begins to bulge locally, and at the apex, an avascular spot, the **stigma**, appears.
- Ovarian contractions extrude the oocyte, which together with its surrounding granulosa cells from the ovary.
- Some of the granulosa cells then rearrange themselves around the zona pellucida to form corona radiata.



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CORPUS LUTEUM

After ovulation, granulosa cells remaining in the wall of the ruptured follicle, together with cells from the theca interna, are vascularized by surrounding vessels. Under the influence of LH, these cells develop a yellowish pigment and change into **lutein cells**, which form the **corpus luteum** and secrete estrogens and **progesterone**. Progesterone prepares the uterine mucosa for implantation of embryo.



OOCYTE TRANSPORTATION

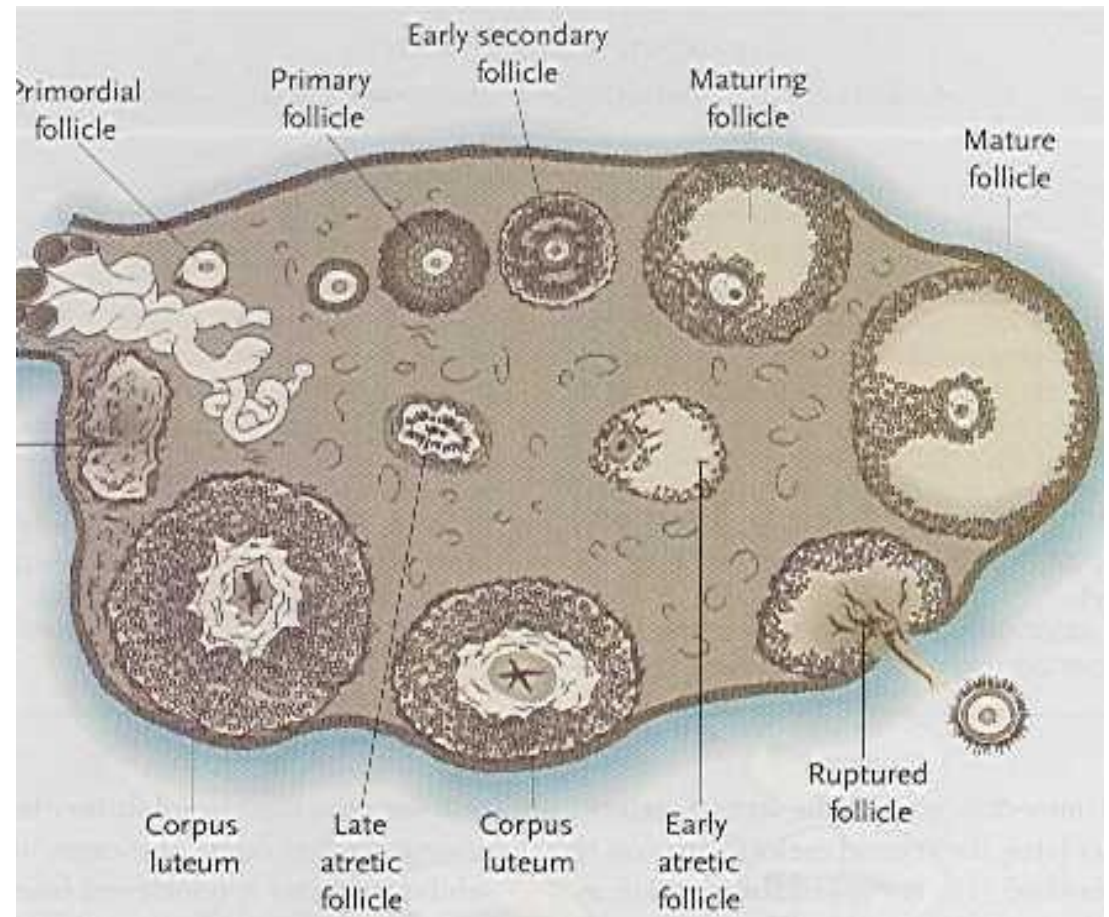
The oocyte, surrounded by corona radiata cells, is carried into the tube by the sweeping movements of the fimbriae and by the motion of cilia on the epithelial lining.

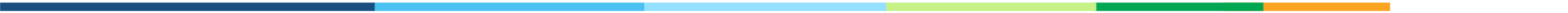
Once the oocyte is in the uterine tube, it is propelled by peristaltic muscular contractions of the tube and by cilia.

In humans, the fertilized oocyte reaches the uterine lumen in approximately 3 to 4 days.

CORPUS ALBICANS

If fertilization does not occur, the corpus luteum shrinks because of degeneration of lutein cells (**luteolysis**) and forms a mass of fibrotic scar tissue, the **corpus albicans**. Simultaneously, progesterone production decreases, precipitating menstrual bleeding.



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- If the oocyte is fertilized, degeneration of the corpus luteum is prevented by **human chorionic gonadotropin**, a hormone secreted by the developing placenta.
 - The corpus luteum continues to grow and forms the **corpus luteum of pregnancy**.
 - Yellowish luteal cells continue to secrete progesterone until the end of the fourth month; thereafter, they regress slowly as secretion of progesterone by the placenta becomes adequate for maintenance of pregnancy.

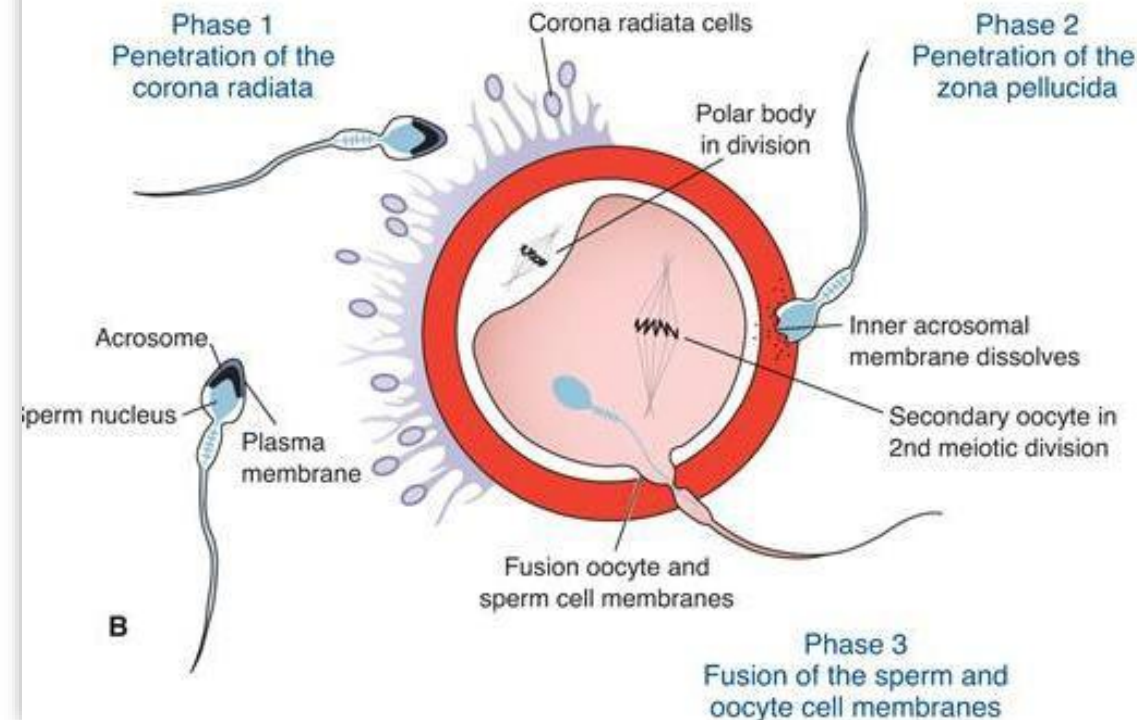
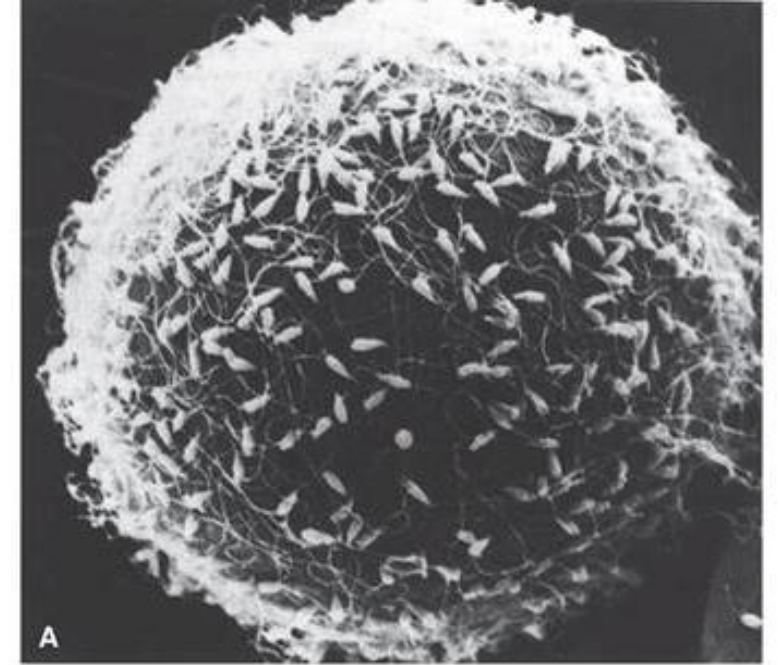
FERTILIZATION

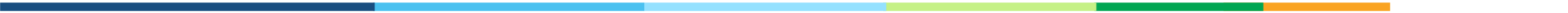
- Fertilization, the process by which male and female gametes fuse, occurs in the **ampullary region of the uterine tube**. This is the widest part of the tube and is close to the ovary.
- Before fertilization the sperm has to undergo **capacitation** and **acrosomal reaction**.
- **Capacitation** is process is a period of conditioning in the female reproductive tract that lasts approximately 7 hours, during which a glycoprotein coat and seminal plasma proteins are removed from the plasma membrane overlying the acrosomal region of the spermatozoa.
- The **acrosomal reaction**, which occurs after binding to the zona pellucida, causes the release of enzymes needed to penetrate the zona pellucida.

PHASES OF FERTILIZATION

The phases of fertilization include the following:

- Phase 1, penetration of the corona radiata
- Phase 2, penetration of the zona pellucida
- Phase 3, fusion of the oocyte and sperm cell membranes





During fertilization, the spermatozoon must penetrate:

1. The corona radiata

2. The zona pellucida

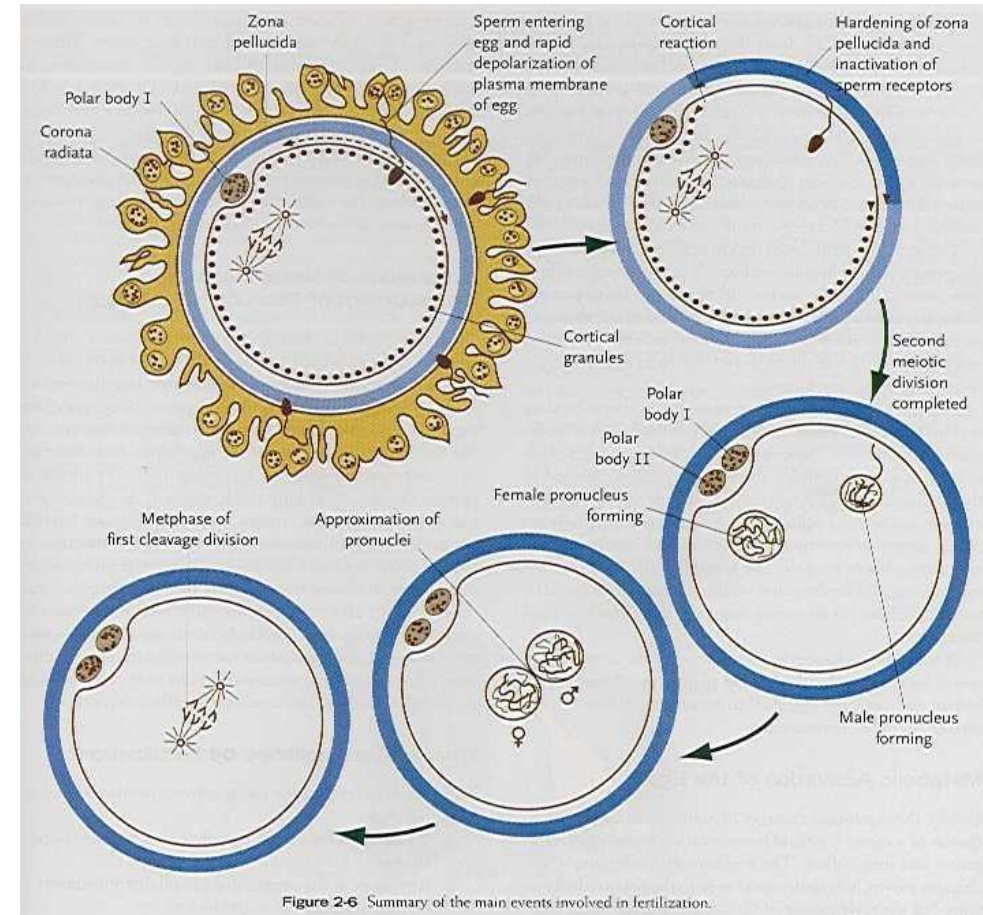
3. The oocyte cell membrane

As soon as the spermatocyte has entered the oocyte:

The oocyte finishes its second meiotic division and forms the **female pronucleus**.

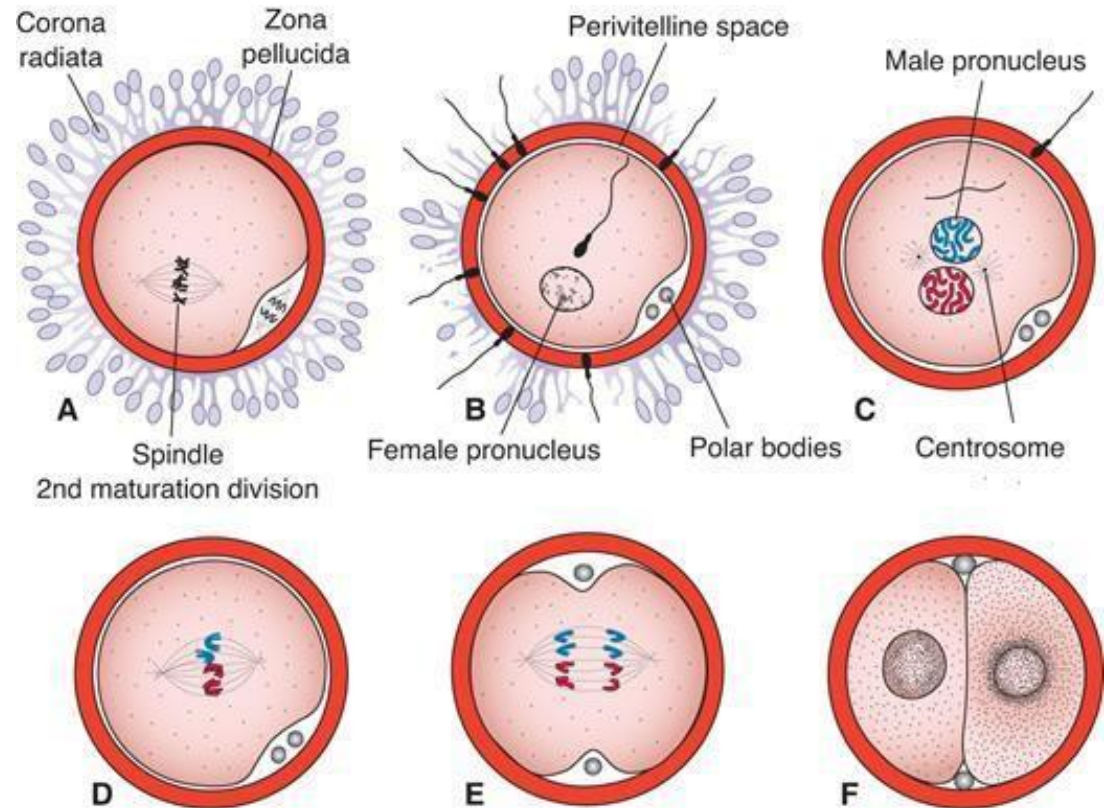
The zona pellucida becomes impenetrable to other spermatozoa.

The head of the sperm separates from the tail, swells, and forms the **male pronucleus**



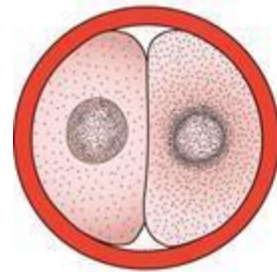
RESULTS OF FERTILIZATION

- **Restoration of the diploid number of chromosomes**
- **Determination of chromosomal sex**
- **Initiation of cleavage**

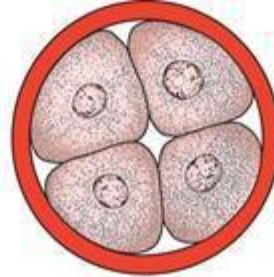


CLEAVAGE

- **Cleavage** is a series of mitotic divisions that results in an increase in cells, **blastomeres**, which become smaller with each division.



Two-cell stage

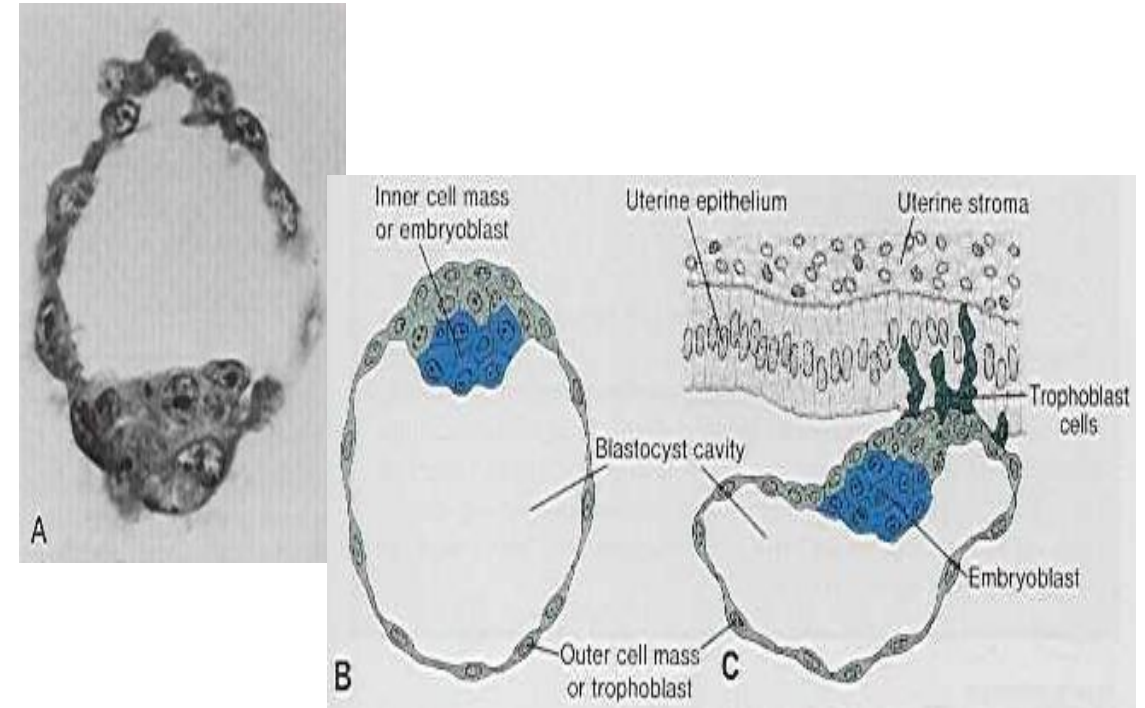


Four-cell stage



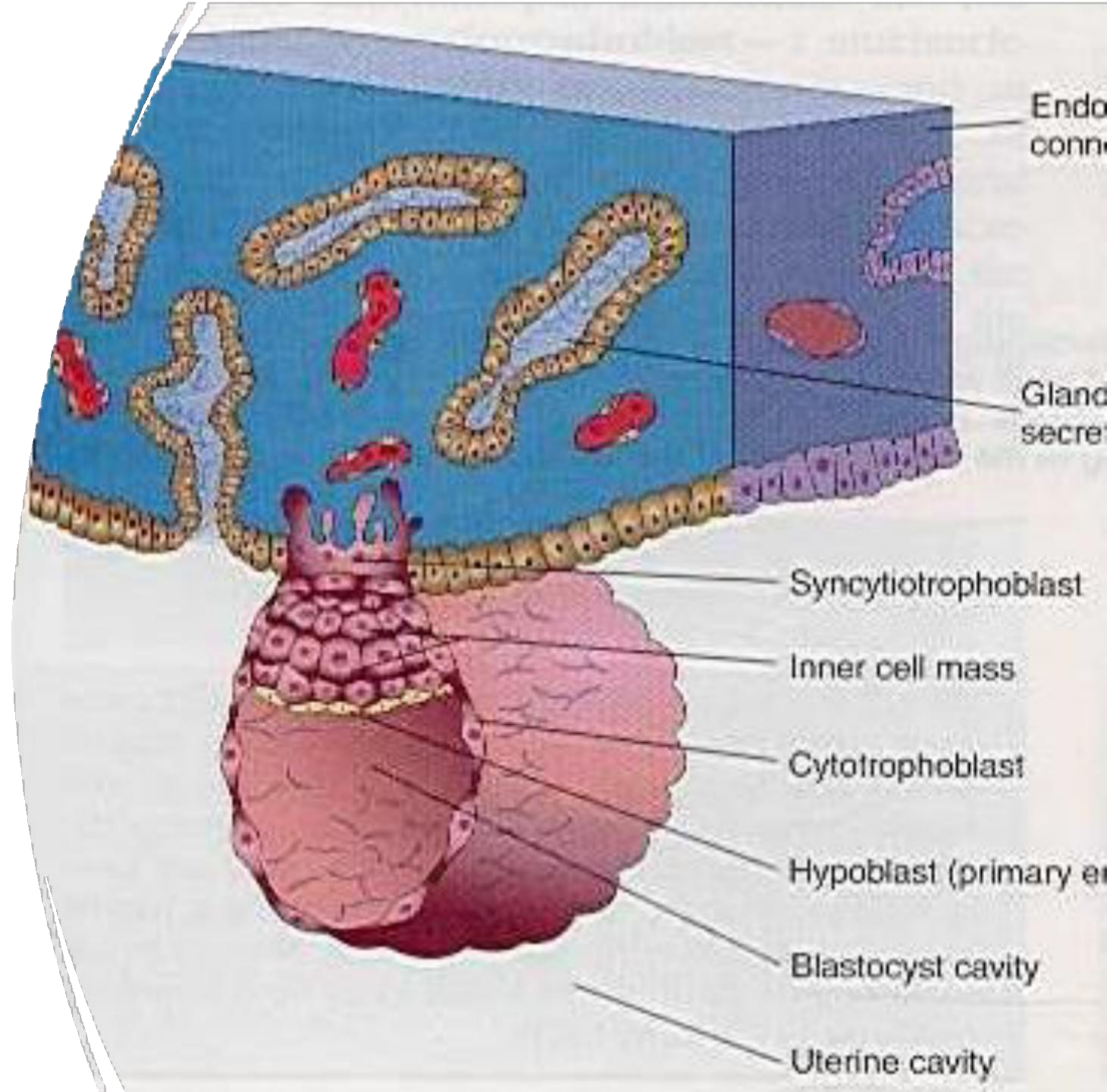
Morula

- After three divisions, blastomeres undergo **compaction** to become a tightly grouped ball of cells with inner and outer layers. Compacted blastomeres divide to form a 16-cell **morula**. As the morula enters the uterus on the third or fourth day after fertilization, a cavity begins to appear, and the **blastocyst** forms.
- Inner cells of the morula constitute the **inner cell mass**, and surrounding cells compose the **outer cell mass**.
- Cells of the inner cell mass, now called the **embryoblast**, are at one pole, and those of the outer cell mass, or **trophoblast** form the epithelial wall of the blastocyst.



IMPLANTATION

- Implantation takes place at the end of first week.
- blastocyst implants in the endometrium along the anterior or posterior wall.



FIRST WEEK OF DEVELOPMENT-SUMMARY

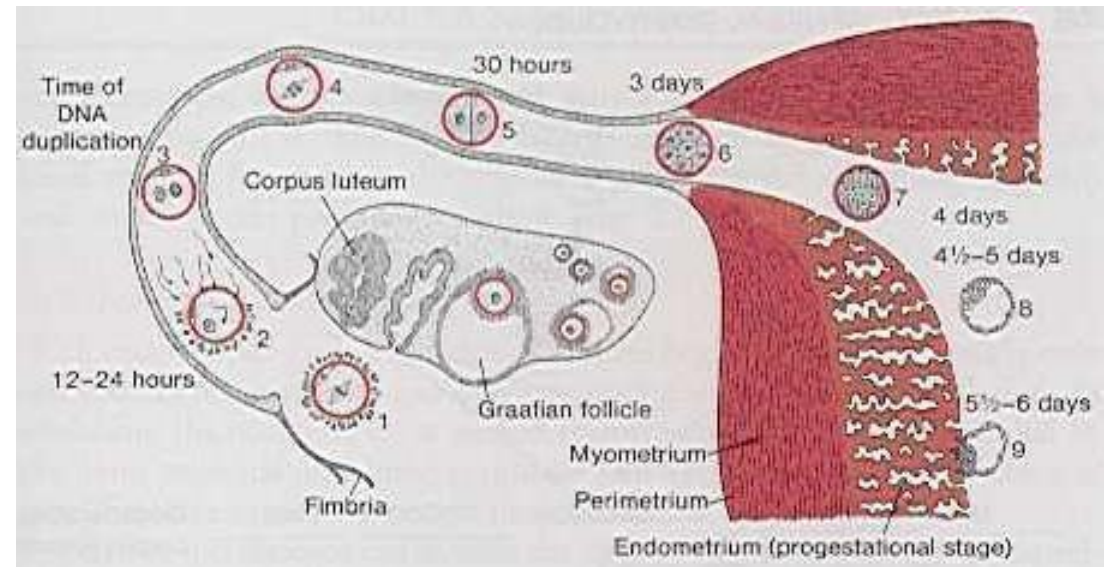
Events include **ovulation** to **implantation**.

Cleavage occurs as the **zygote** passes down the uterine tube.

Approx. 3 days after **fertilization**, the 16-cell **morula** is formed.

Blastocyst formation:

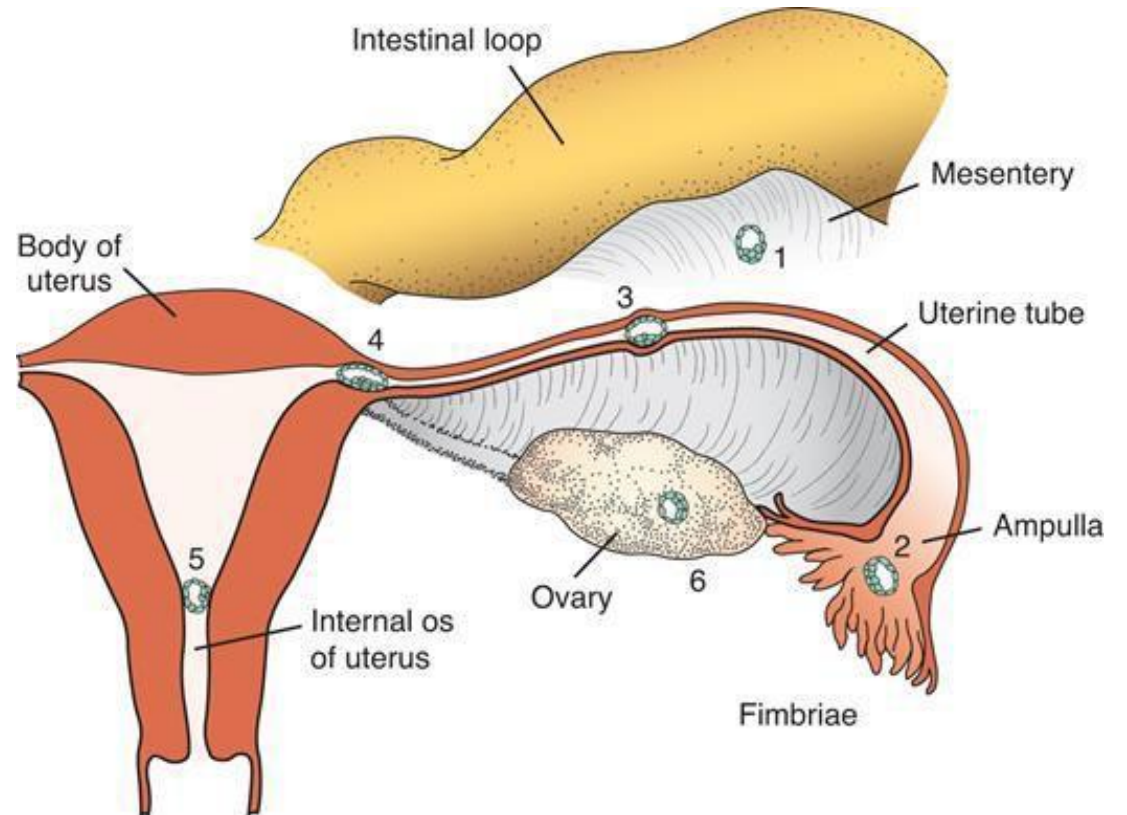
- **embryoblast**
- **trophoblast**



ECTOPIC PREGNANCY

Ectopic pregnancies may occur at any place in the abdominal cavity, ovary, or uterine tube (Fig. 4.8). Ninety-five percent of ectopic pregnancies occur in the uterine tube (80%), abdominal cavity or in the ovary.

Ectopic pregnancies occur in 2% of all pregnancies and account for 9% of all pregnancy-related deaths for the mother.



References

1. Langman's medical embryology by Saddler, T. W., 15th ed., 2024, Wolters Kluwer.
2. Before we are born: Essentials of embryology and birth defects by Moore, K. L., Persaud, T. V. N., & Torchia, M. G., 9th ed., 2016, Elsevier